AeroLab

How to play

Version 0.2: multi-stage rockets, flight programs and orbit

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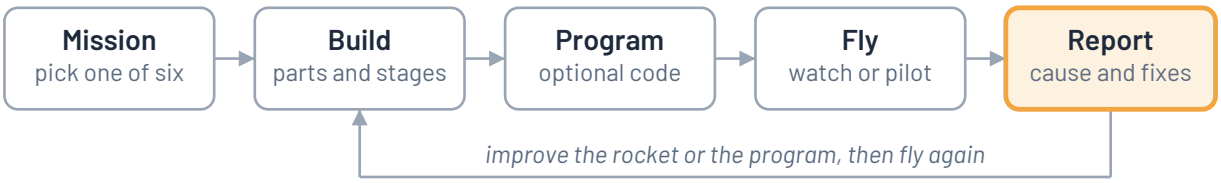
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1 Welcome

AeroLab is a rocket engineering game where the physics decides: a rocket reaches orbit only when its design, and its flight program if you write one, are both right. Nothing is scripted. Every number you see comes from the same simulation, so the builder, the flight and the report always agree.

Every flight ends in a report that tells you what to change



You will fail often, and that is the point. Each failure report names the cause, shows the evidence and suggests what to change, so the next flight goes further.

The six missions climb from a 10 km hop to a circular orbit flown entirely by your own code. You need no rocket science to start; section 3 covers the five ideas that explain almost every result.

2 Getting started

Open `index.html` from the project folder in Chrome, Edge, Firefox or Safari. Nothing needs installing and it runs offline. You can also play online at the published link, although that page may block flight programs; manual flying still works there.

The top bar has two tabs and a mission menu:

Screen	What you do there
Build tab	Assemble the rocket, read its engineering numbers and fix the problems the checks list
Program tab	Write or load a flight program, check it and fly the mission instantly
Launch pad and flight	Opened with **Go to launch pad** at the bottom right; fly by hand or watch your program
Mission menu (top right)	Choose the mission. A tick marks the ones you have completed

The bottom bar is always there: name your design, **Save version**, **Load design**, and **Go to launch pad**. The launch button stays disabled until the checks show no errors, and the text beside it tells you whether the flight program is on.

Your first five minutes

1. Keep mission **M002: Edge of space** selected.
2. On the Build tab, choose **Sounder** from *Start from a reference rocket*.
3. Press **Go to launch pad**, then Space to ignite.
4. Press **.** a few times to speed up time, and watch the payload pass 100 km.
5. Read the report, then try removing the fins on the Build tab and flying again. Section 8 explains what happens.

3 Rocket science in five ideas

Five numbers explain nearly every success and failure in AeroLab. The builder shows all of them before you launch.

Idea	What it means	Rule of thumb in AeroLab
Thrust-to-weight (TWR)	Engine thrust divided by the rocket's weight	Below 1.0 it never leaves the pad. 1.5 to 3.5 is healthy. Far above that, it goes too fast in thick air
ΔV (delta-v)	The total speed change the propellant can give, from the rocket equation	About 2,500 m/s reaches 100 km. Orbit needs about 9,000 m/s, which is why orbital rockets have stages
Stability margin	How far the centre of pressure (CP) sits behind the centre of gravity (CG), in body diameters (calibres)	Positive means stable, like an arrow. 1 to 3 calibres is typical. Negative means it flips unless the engine steers it
Dynamic pressure (q)	The force of the airflow, ½ × air density × speed^2	Tanks and payloads fail above 120 kPa. Flying at an angle to the airflow in thick air breaks the rocket even sooner
Orbit	Falling around Earth fast enough to keep missing it	At 200 km you need about 7,800 m/s sideways. The periapsis (lowest point) must be above about 150 km to stay up

Two more words appear everywhere. **Apoapsis** is the highest point of your current path and **periapsis** the lowest. Burning forward at apoapsis raises periapsis, and that is how every orbit in this game is made.

4 Building a rocket

A rocket is a stack of parts from nose to engine; click a part in the list on the left and it drops into the right place.

Where parts go

- Nose cones always go on top, replacing any nose already there.
- Payload, flight computer and reaction wheels go into the top stage.
- Tanks and engines go into the stage picked under *Tanks and engines go into*. A new engine replaces the engine already in that stage.
- Fins attach to the tail of stage 1 and fall away with it.
- Use ▲ and ▼ in *Your stack* to reorder, and ✕ to remove. Click any part, in the stack or the drawing, to read about it below the drawing.

Reading the drawing. The filled blue circle is the centre of gravity with full tanks, the hollow one with empty tanks. The amber ring is the centre of pressure. For a stable rocket the amber ring sits below both blue circles. Dashed amber lines mark where the stages separate.

The engineering panel lists lift-off mass, TWR, total ΔV, length, CG, CP, stability margin and the structural limit. Green means healthy, amber means caution, red means a problem. The ΔV bar has a mark showing roughly what the selected mission needs. The stage table below it gives each stage's ΔV, TWR, burn time and mass.

The checks list decides whether you can launch. Red errors block the launch button, and each one says how to fix it. Amber warnings let you fly, but usually predict how the flight will end.

Building a second stage

1. Build the upper stage first: nose, payload, flight computer, a tank and an engine.
2. Press **Add a stage below**. A decoupler appears at the bottom, starting a new, empty stage 1, and the old stack becomes stage 2.
3. With *Stage 1* selected, add tanks and an engine. They go below the decoupler.
4. Add fins if the stability margin is negative.

The rules the checks enforce for multi-stage rockets:

- Every stage except the top one needs an engine at its bottom and a tank in the same stage.
- The flight computer and the payload must be in the top stage, or they would be thrown away.
- The upper stage needs reaction wheels if it must turn while its engine is off, for example to line up for an orbit burn. Without them you get a warning, not an error.
- Some missions demand at least two stages.

5 Flying by hand

On the pad, press Space to ignite; the rocket lifts off once thrust beats its weight, and attitude hold keeps it pointed straight up.

Key	Button	Action
Space	Ignite	Light the current stage's engine
G	Stage	Separate the spent stage
W / S	Throttle slider	Throttle up or down, 40 to 100%, liquid engines only
A / D or ← / →	◀ Pitch / Pitch ▶	Tilt west or east. With *Hold pitch* on, this moves the held angle
X	Cut engine	Shut down a liquid engine
M	Map	Switch to the map view; scroll to zoom in on the rocket
N	Sound on / off	Mute or unmute; the choice is remembered
, / .	1× to 100×	Slow down or speed up time
Esc	End flight	Stop and open the report

The attitude menu chooses what the rocket points at:

- *Hold pitch*: a fixed angle above the horizon. 90° is straight up.
- *Hold surface prograde*: along the airflow, at zero angle of attack. Use it in the atmosphere.
- *Hold orbital prograde*: along your orbital velocity. Use it above about 70 km, for orbit burns.
- *Hold off*: no help. The pitch keys swing the engine directly.

What the screen shows

- Top left: the mission clock, altitude and live readouts, including speed, dynamic pressure with a bar against the limit, angle of attack, thrust, TWR, stage propellant, apoapsis and periapsis.
- Top centre: the mission objective and your progress towards it.
- Top right: which stage is firing and its engine state, the flight program's status, and a small altitude-against-downrange plot. Stage 1 is drawn in amber and upper stages in blue.
- Right: the event log, with every ignition, separation, max Q and failure time-stamped.

Staging by hand. Wait for *Stage 1: Propellant depleted, engine cut-off* in the log. Press G to separate, then Space to light stage 2 straight away. A finless upper stage needs its engine to steer while it is still in the air. Pressing G while the engine is still firing destroys both stages.

Time warp never changes the result. The simulation always takes the same 0.01 s steps and simply runs more of them per second.

Sound and effects. Listen as well as watch. The engine roar fades to a muffled rumble as the air thins, and wind noise rises with dynamic pressure. Ignition, separation, cut-off and going supersonic each have their own sound. A two-tone alarm and a blinking *Master caution* badge warn when you pass 85% of the structural, bending or payload g limit, so ease off before something breaks. Crashes explode with a fireball, sparks, smoke and the rocket's own parts tumbling apart, and raise a dust cloud on the ground. Browsers only start sound after your first click or key press. Time warp above 10× mutes the engine.

The map view draws Earth, the path you have flown, your predicted orbit (dashed), and markers for apoapsis and periapsis, all to true scale. A 200 km orbit really is that thin ring around the planet.

6 Programming the flight computer

A flight program is JavaScript that flies the rocket for you, and the quickest way in is to load an example on the Program tab and press **Test flight**.

The Program tab

- **Examples** (left): six programs, each loaded together with the rocket and mission it was written for.
- **Editor** (centre): line numbers turn red on the line with an error. Tab indents, and Ctrl+Enter (Cmd+Enter on a Mac) runs Check.
- **Check**: compiles the program and reports syntax errors with their line.
- **Test flight**: flies the whole mission instantly and shows the verdict, key numbers, the event log and a trajectory sketch. **Watch this flight** then launches it for real; **Full report** opens the report.
- **Fly with this program**: when ticked, the launch pad hands control to the program. Untick it to fly by hand.
- **The ship object** (right): every command and reading your program can use.

A first program

```
function* flight(ship) {
  ship.throttle(1);       // full throttle
  ship.pitch(90);         // point straight up
  ship.ignite();
  yield ship.until(() => ship.propellant <= 0);   // wait for burnout
  ship.log('Burnout at', ship.altitude / 1000, 'km');
}
```

The asterisk in `function*` lets the program wait. It runs 20 times per simulated second, and each `yield` hands control back to the simulation:

Write	It waits
`yield;`	One tick, 1/20 s. Use it inside steering loops
`yield ship.wait(5);`	5 simulated seconds
`yield ship.until(() => ship.altitude > 10000);`	Until the condition is true

Three rules

1. Every loop that waits must contain a `yield`, or the computer halts the program for running too long.
2. `ship.stage()` only separates. Light the next stage with `ship.ignite()`.
3. In the air, steer with `ship.surfacePrograde()`. Switch to `ship.prograde()` only above the atmosphere.

During a programmed flight the pad shows *Flight program loaded*; press Space to start it. The manual controls are locked while the program flies, and the program box at the top right shows what it is waiting for. **Take manual control** stops the program and hands you the controls, but it counts as manual input. If the program crashes, the rocket carries on with its last commands, and the report names the line.

The same program on the same rocket always flies exactly the same way. The full command list and a phase-by-phase walk to orbit are in `docs/PROGRAMMING_GUIDE.md` in the project folder.

7 Reading the flight report

The report opens when a flight ends, or when you press End flight or View flight report; it tells you what happened and what to change.

Part of the report	What it tells you
Verdict and cause	*Mission complete* or *Mission failed*, and the one root cause in a sentence
Evidence	The numbers behind the cause: altitudes, pressures, ΔV available, losses to gravity and drag
Contributing factors	Why it happened, for example an unstable design, a stage separated with fuel left, or a program error
Staging	Each separation's time, altitude, dynamic pressure and leftover propellant
Flight program	Whether it ran to the end or halted, and on which line
What to try, or Next	Concrete fixes, or the next challenge after a success
Stat tiles	Peak altitude, max Q, peak g and Mach; for orbit missions, periapsis, apoapsis and peak orbital speed
Telemetry chart	Pick a channel: altitude, speeds, apoapsis, periapsis, acceleration, dynamic pressure, thrust, mass or pitch. Dashed lines mark targets and limits, and stage separations are marked. Hover to read values

The buttons underneath: **Edit program** jumps to the Program tab and highlights the error line, **Fly again** repeats the flight, and **Back to builder** returns to the Build tab.

Two habits pay off. Read the cause before changing anything, because the obvious fix is often wrong: a rocket that fell short may need less drag, not more fuel. And change one thing at a time, so the next report tells you whether it worked.

8 Mission walkthroughs

Each mission adds one idea; every result below was flown and checked in the game's own tests.

Mission	Objective	Reference rocket	Needs
M001 First flight	Payload above 10 km	Sounder	Any rocket that flies
M002 Edge of space	Payload above 100 km	Sounder	Stability, about 2,500 m/s
M003 Two stages	Two or more stages, payload above 500 km	Twin	Clean staging
M004 Orbit	Periapsis above 150 km, engine off	Orbiter	About 9,000 m/s and guidance
M005 Circular orbit	Periapsis and apoapsis both 180 to 220 km	Orbiter	A timed circularisation burn
M006 Hands off	M004 with no manual input	Orbiter	A complete flight program

M001: First flight

Load the Sounder, launch and ignite. Almost anything with a liquid engine and a flight computer clears 10 km. Even the finless starter rocket completes the objective before it breaks up at about 35 km.

M002: Edge of space

The Sounder, with a Sparrow engine, tank M and small fins, reaches 134.5 km. Try these variations to see the ideas from section 3:

- **Remove the fins.** The gimbal holds it straight while the engine runs, then it breaks up at burnout, around 35 km. The CP sits ahead of the CG.
- **Large fins** still succeed at 111.5 km. More drag costs altitude.
- **Tank S** falls short at 20 km. Not enough ΔV.
- **Heavy payload** needs tank L: 85 km with tank M, 168 km with tank L.
- **The Heron engine at full throttle** overloads the airframe at 4 km. Throttle it to about 45% between 3 and 20 km and it reaches 214 km. The *Throttle through max Q* example program does this automatically.
- **The solid motor** tumbles without fins and falls short with small fins, but reaches 121 km with large fins. It cannot steer, so only fins keep it straight.

M003: Two stages

The Twin puts a Sparrow upper stage on a Heron first stage with two L tanks. Load the *Two-stage staging* example and it reaches about 950 km, separating at 35 km where the air is thin. To fly it by hand, wait for stage 1 burnout, press G, then Space at once.

What goes wrong: separate while stage 1 is firing and it rams the upper stage. Forget to light stage 2 and the finless stage tumbles in the air. A version with less stage 1 propellant separates at 28 km, where the dynamic pressure is about 16 kPa; it only survives with small fins on stage 1.

M004: Orbit

The Orbiter has a Condor first stage with tanks C5, C5 and C3, and a Kestrel vacuum upper stage with a C5 tank and reaction wheels: 9,567 m/s in all. Load the example *Orbit: gravity turn, coast, circularise* and press Test flight. It reaches a 193 × 200 km orbit in under seven minutes, with about 30 kg of propellant to spare.

The program does five things, in order:

1. Climb, then tilt east gradually, ending at 30° above the horizon by 50 km.
2. Separate at burnout, around 55 km, and point along the airflow.
3. Burn the upper stage until the apoapsis reaches 200 km, then cut off.
4. Coast towards apoapsis, turning prograde with the reaction wheels.
5. Relight at apoapsis and burn until the periapsis rises to meet it.

Flying this by hand is hard but possible. Watch the Apoapsis and Periapsis readouts, and use *Hold orbital prograde* for the final burn.

M005: Circular orbit

The same example program lands inside the 180 to 220 km band, because it times the burn to be centred on apoapsis. The *direct insertion* example also qualifies, at about 190 × 197 km. If your orbit is too elliptical, cut the engine sooner once the periapsis is high enough.

M006: Hands off

Any orbit program that finishes the job works. On the pad press Space to start it, then do nothing but watch. Time warp and the map view are allowed; ignite, stage, throttle, pitch, the attitude menu and Take manual control all count as manual input and fail the mission instantly.

9 Troubleshooting

Most failures fall into a handful of patterns, and the report's cause line tells you which one you hit.

Report says	Usual cause	Fix
Never left the pad	TWR below 1.0	A stronger engine, or less mass
Structural overload	Too much thrust in thick air	Throttle down between about 5 and 15 km, or use a smaller engine
Aerodynamic break-up	Flying sideways into dense air, often an unstable design at burnout	Add fins; turn gradually; keep attitude hold on

Report says	Usual cause	Fix
Loss of control, tumbled	CP ahead of CG, or a finless upper stage in the air	Larger fins; separate higher; light the upper stage at once
Payload destroyed by g	A light, nearly empty rocket still at full thrust	Throttle down late in the burn
Recontact	Staged while the lower engine was still firing	Wait for burnout or cut the engine, then stage
Stage 2 was never ignited	Staged but forgot to light the next engine	Press Space after G, or call `ship.ignite()` after `ship.stage()`
Insufficient ΔV	Not enough propellant, or large gravity and drag losses	Bigger tanks, a lighter payload, or another stage
Did not reach orbit	Too little sideways speed, or no circularisation burn	Turn further toward the horizon; burn at apoapsis to lift the periapsis
Orbit but not in the band	The circularisation burn ran too long or started off-centre	Cut off once periapsis is in range; centre the burn on apoapsis
Program halted	A code error, or a loop with no `yield`	Press Edit program; the line is highlighted
Manual input during a hands-off mission	Touched the controls in M006	Let the program do everything; time warp and the map are allowed

Other problems

- *The launch button is greyed out:* the checks list has a red error. Fix it and the button unlocks.
- *Launching jumps to the Program tab: Fly with this program* is ticked, but the program has an error. Fix the program or untick the box.
- *Keys do nothing in the Streamlit dashboard:* click inside the game once so it receives the key presses.
- *The Program tab says custom code is blocked:* the online page's security settings forbid it. Open `index.html` from the project folder instead.
- *My saved designs are missing:* saves live in the browser you used. The standalone game and the Streamlit Play page keep separate saves.

10 Quick reference

Flight keys: Space ignite, G stage, W/S throttle, A/D pitch, X cut engine, M map, N sound, comma and full stop for time warp, Esc end flight (full table in section 5).

Parts

Part	Mass	Key figures	Best for
Ogive nose cone	18 kg	Low drag, 1.2 m	Every rocket
Conical nose cone	12 kg	Lighter, more drag	Saving mass
Science payload	25 kg	10 kg instrument, survives 15 g	All missions
Heavy payload	85 kg	60 kg instrument, survives 12 g	A harder M002

Part	Mass	Key figures	Best for
Flight computer	12 kg	Runs the program, steers every engine	Every liquid-fuelled rocket; top stage
Reaction wheels	9 kg	400 N·m, turns without thrust	Orbital upper stages
Tank S / M / L	40 / 70 / 100 kg empty	160 / 330 / 500 kg propellant	Sounding rockets
Carbon tank C3 / C5	38 / 60 kg empty	540 / 900 kg propellant	Orbital stages
Stage decoupler	15 kg	600 N·s separation push	Multi-stage rockets
LR-12 Sparrow	55 kg	16 kN, Isp 250 to 285 s, 1 ignition	Small first or upper stages
LR-40 Heron	110 kg	42 kN, Isp 262 to 298 s, 1 ignition	Big sounding rockets, Twin stage 1
LR-60 Condor	125 kg	68 kN, Isp 284 to 312 s, 1 ignition	Orbital first stages
RV-9 Kestrel	30 kg	12 kN, Isp 170 to 345 s, 3 ignitions	Upper stages in vacuum only
SR-9 Needle solid motor	740 kg loaded	24 kN, no throttle, steering or shutdown	Simple boosters, with large fins
Small / large fins	10 / 20 kg	Moves the CP aft; small or strong effect	Stability on stage 1

Thrust figures are vacuum ratings. The first Isp number is at sea level, the second in vacuum.

Saving. *Save version* stores the rocket, its flight program and its mission as a new numbered version, such as Orbiter_v001, and never overwrites an older one. *Load design* brings any version back. Your last design and program are restored automatically when you reopen the game, and V0.1 saves still load.

Going further. The Streamlit dashboard, started with F5 in VS Code, flies rockets headlessly and sweeps a program constant across many flights. For example, it shows which gravity-turn angles reach orbit. The README explains the setup, and `docs/PROGRAMMING_GUIDE.md` covers every command.